MEMENTO: A MOBILE APPLICATION FOR NOTICING AND PRESERVING MEMORIES

September 6, 2026

PROJECT PLAN

1. Project Summary

Everyone has moments that they would have liked to remember in more detail, but we often forget to record these everyday experiences. Our phones already keep track of a lot of information about us, so why not use yours to notice these moments for you? Journaling apps tend to rely on the user to remember to open the app, while social media can be too public and try to keep the user scrolling as long as possible. Using your phone sensors all while keeping your privacy intact, Memento will detect novel experiences and remind you to create a memory. Users can add things like notes, photos, and more to their memories and save them for themselves or share them with close friends and family.

2. Project Scope

Memento is a journaling and scrapbook application that will only use information that the user has permitted, such as location, weather, activity, or motion, to identify **Vignette Candidates**: moments in the user's day that may be noteworthy. Determining which experiences are meaningful is subjective, so the app will not decide which ones to save. Instead, these candidates will later be presented to the user for review, allowing them to choose which they would like to save, after which any unused data will be discarded.

The project includes account creation and authentication, permission selection during onboarding, configuration of a Routine or Home Zone, customizable notifications for Vignette Candidates, detection and temporary storage of Vignette Candidates, and a daily reflection interface where users can review detected moments or manually add one that was missed. Users will also be able to manually create these Vignettes by attaching selected information such as photos, written reflections, mood, general location, weather, activity, as well as date and time. These will be saved in a Scrapbook interface for viewing and organizing saved memories.

Users who add each other as friends can choose to turn their Vignettes into Postcards so that they may share them with each other as well. However, Memento is not intended to be a social media platform, so many features related to such a platform, such as follower count, reposting, or public content sharing, to share only those, are outside its scope.

Potential stretch goals that are not in our initial scope include more sophisticated novelty scoring (potentially with machine learning), smartwatch integration, monthly Scrapbook retrospectives, and additional social interactions.

3. Project Team

Judhy Germain – Group Leader
Cameron Lowe
Paul Bagaric

4. Stakeholders

The main stakeholders would be the individual users as well as whichever friends and family members of theirs who are also using the app. However, another important one to consider would be the parents of minors who might be worried about their children sharing content with the wrong person. Although it is in the user's hands what they share and with whom, there could be a worry that a child may befriend someone on the app and share with them inappropriate content. This is something that, should we ever release the app to the public, we could address through parental controls of some kind.

5. Technical Description

A collection of all the data points the user exposes to the app on setup will be stored locally. This information will be processed on the device to identify possible moment candidates based on factors such as location novelty, distance from the Home Zone, elevation changes, and weather. Raw sensor and GPS data will be temporarily stored and deleted after processing. Moment detection will initially use a simple rule-based scoring system. Related detections will be grouped together so that one outing does not generate multiple candidates.

At the user's chosen reflection time, the app will send a notification and display detected Moment Candidates. The user can save a candidate, ignore it, or manually create a memory. Saved memories become Vignettes and may include additional information such as photos, text, mood, location, date, and time. After reflection time is over, all information about unsaved moments will be deleted.

A backend service will handle account management, friend relationships, and sharing. Vignettes will remain visible only to their owner unless deliberately shared by turning them into a Postcard. When sending a Postcard, the user will control which fields friends are allowed to see.

6. Project Standards

Memento will be developed in Android Studio Quail 4 (2026.1.4) using Flutter 3.47.2 and Dart 3.13.2. The application will target Android SDK 37 and use OpenJDK 25.0.3. Dart will be the application's primary programming language, but we may use Kotlin if there is an Android-specific functionality we need that requires it. Flutter will also be used for automated mobile tests while JEST will be used for API and backend testing.

To store data locally on the device, we will use SQLite. We will use a MySQL database as well as DigitalOcean Spaces for anything that requires being stored on a server. The latter will be hosted on DigitalOcean directly and managed through REST APIs built with Node.js and Express. The APIs will use JSON over HTTPS and will be documented and tested through SwaggerHub. We will also use external APIs, such as SendGrid for email verification and Open-Meteo for weather tracking.

Figma will be used to produce our wireframes and mockups, GitHub will be used for version control and repository hosting, Jira will be used for task management and the Gantt chart, and Discord will be used for team communications.

7. Software Features

Users can create an account, verify their email, sign-in and, and manage basic account information. During onboarding, permissions will be given to use contextual data such as location, weather, activity, motion, and elevation. A Routine/Home Zone can be defined so that familiar locations are not treated as novel. The application can analyze these permitted signals to identify possible memorable experiences, such as traveling far from the Home Zone, unusual movement, etc. At a user-selected time, Memento can notify the user and present detected Vignette Candidates for review. Another option is manual memory creation, for when the user thinks of their own idea. Vignette Candidates can be saved and edited to add photos, text reflections, mood, location, weather, activity, and date/time information. The Scrapbook allows users to browse their saved Vignettes and filter them by Vignettes, Postcards, Favorites, and Albums. Vignettes can be added to favorites or grouped into albums. They can add friends and deliberately share Vignettes with them, in which case the Vignette becomes known as a Postcard. When editing a Postcard, privacy controls can be edited to control which friends can see it, and which fields they are allowed to see.

8. Risk Mitigation Plan

Immediate risks include security in shepherding customer data from their sensors to our app, then from our app to our server, and back. We'll mitigate this by using industry-standard security technologies. As well, any kind of user-generated content at any scale will require moderation efforts to prevent bad actors abusing the platform and exposing us to moral and legal liability. At launch scale we would invest in a professional moderation team; in the development stages we'll simply restrict our userbase to trusted individuals.

Each individual type of data that is collected will have a permission associated with it during onboarding when the user first opens the app. Additionally, raw sensor data will be deleted when no longer needed, and Vignette Candidates are deleted once the user chooses which ones they actually want to save. Additionally, communication with the backend server will use encryption, passwords will be securely hashed, and access controls will prevent unauthorized users from viewing private memories.

9. Deliverables

Over the course of this project, we will produce the following deliverables:

- Planning Documentation – September 6th, 2026 (this document)
- Stakeholder Interviews – September 13th, 2026
- Design Documentation – September 20th, 2026
- Initial UI Implementation – October 25th, 2026
- Full application – November 16th, 2026
- User Evaluations – December 6th, 2026

For the initial UI implementation, we will specifically work on implementing our design, but the app will not fully be functional until the full application release three weeks later. It is during those three weeks that we will work on the application's back-end development.

10. Schedule & Milestones

This is our complete task list, including sub-tasks. For our Gantt chart, see *Memento - Gantt Chart.pdf*.

Work	Assignee	Priority	Status	Resolution	Start date	Due date
~~MEM-1~~ Setup	JG Judhy Germain-Bouchard	Highest	Done	Done	Aug 31, 2026	Sep 2, 2026
~~MEM-7~~ Create a GitHub repo	JG Judhy Germain-Bouchard	Highest	Done	Done	Sep 2, 2026	Sep 2, 2026
~~MEM-2~~ Initialize flutter project	JG Judhy Germain-Bouchard	Highest	Done	Done	Sep 2, 2026	Sep 2, 2026
~~MEM-8~~ Create Discord Server	JG Judhy Germain-Bouchard	Low	Done	Done	Aug 31, 2026	Aug 31, 2026
~~MEM-9~~ Provide Workspace Setup Instructions	JG Judhy Germain-Bouchard	Medium	Done	Done	Sep 2, 2026	Sep 2, 2026
MEM-5 Test	Unassigned	High	To Do	Unresolved	Sep 28, 2026	Nov 29, 2026
MEM-90 User Evaluation	Unassigned	Highest	To Do	Unresolved	Nov 16, 2026	Nov 24, 2026
MEM-85 Prepare Test Plan	Unassigned	High	To Do	Unresolved	Oct 25, 2026	Nov 1, 2026
MEM-88 Conduct Testing	Unassigned	Highest	To Do	Unresolved	Nov 9, 2026	Nov 29, 2026
MEM-89 Device Testing	Unassigned	Highest	To Do	Unresolved	Nov 16, 2026	Nov 29, 2026
MEM-35 Unit Testing	Unassigned	Highest	To Do	Unresolved	Sep 28, 2026	Nov 16, 2026
MEM-86 Functional and Integration Testing	Unassigned	Highest	To Do	Unresolved	Nov 9, 2026	Nov 22, 2026
MEM-6 Deliver	Unassigned	Highest	To Do	Unresolved	Nov 16, 2026	Dec 6, 2026
MEM-92 Submit Project	Unassigned	Highest	To Do	Unresolved	Dec 6, 2026	Dec 6, 2026
MEM-91 Prepare Final Documentation	Unassigned	Highest	To Do	Unresolved	Nov 16, 2026	Dec 6, 2026

Work	Assignee	Priority	Status	Resolution	Start date	Due date
MEM-3 Design	Unassigned	High	To Do	Unresolved	Sep 14, 2026	Sep 20, 2026
MEM-55 Define API Structure	Unassigned	High	To Do	Unresolved	Sep 17, 2026	Sep 20, 2026
MEM-54 Map Navigation and User Flows	Unassigned	High	To Do	Unresolved	Sep 14, 2026	Sep 18, 2026
MEM-50 Decide on Data Storage and Retention	Unassigned	Highest	To Do	Unresolved	Sep 14, 2026	Sep 18, 2026
MEM-51 List Information to be Stored	Unassigned	High	To Do	Unresolved	Sep 14, 2026	Sep 15, 2026
MEM-52 Classify On-device and Server Data	Unassigned	Highest	To Do	Unresolved	Sep 14, 2026	Sep 15, 2026
MEM-53 Define Retention and Clean-Up Rules	Unassigned	Highest	To Do	Unresolved	Sep 15, 2026	Sep 18, 2026
MEM-20 Create App Wireframe	Unassigned	Highest	To Do	Unresolved	Sep 14, 2026	Sep 17, 2026
MEM-21 Entity Relationship Diagram	Unassigned	Medium	To Do	Unresolved	Sep 16, 2026	Sep 20, 2026
MEM-22 Use Case Diagram	Unassigned	Medium	To Do	Unresolved	Sep 14, 2026	Sep 18, 2026
MEM-23 Class Diagram	Unassigned	Medium	To Do	Unresolved	Sep 16, 2026	Sep 19, 2026
MEM-24 Color Palette	Unassigned	Low	To Do	Unresolved	Sep 14, 2026	Sep 15, 2026
MEM-10 Plan	Unassigned	High	In Progress	Unresolved	Sep 5, 2026	Sep 13, 2026
~~MEM-18~~ Identify Milestones and Prepare Schedule	Judhy Germain-Bouchard	High	Done	Done	Sep 5, 2026	Sep 6, 2026
~~MEM-17~~ Identify Potential Risks and Solutions	Cam L	Medium	Done	Done	Sep 5, 2026	Sep 6, 2026
~~MEM-16~~ Plan Software Features	Cam L	Highest	Done	Done	Sep 5, 2026	Sep 6, 2026
~~MEM-15~~ Decide Tech Stack	bagaricpaul	High	Done	Done	Sep 5, 2026	Sep 6, 2026
~~MEM-14~~ Define Solution Architecture and Functionality	bagaricpaul	Highest	Done	Done	Sep 5, 2026	Sep 6, 2026
~~MEM-13~~ Identify Stakeholders	Cam L	Medium	Done	Done	Sep 5, 2026	Sep 6, 2026
~~MEM-12~~ Define the Project Scope	bagaricpaul	Highest	Done	Done	Sep 5, 2026	Sep 6, 2026
~~MEM-11~~ Identify Problem Statement and Proposed Solution	Cam L	High	Done	Done	Sep 5, 2026	Sep 6, 2026
MEM-19 Stakeholder Interviews	Unassigned	Highest	To Do	Unresolved	Sep 8, 2026	Sep 13, 2026

Work	Assignee	Priority	Status	Resolution	Start date	Due date
MEM-4 Front-End Development	Unassigned	Highest	To Do	Unresolved	Sep 21, 2026	Nov 16, 2026
MEM-83 Notifications	Unassigned	High	To Do	Unresolved	Oct 1, 2026	Oct 18, 2026
MEM-36 Documentation	Unassigned	Medium	To Do	Unresolved	Sep 21, 2026	Nov 16, 2026
MEM-26 Code Pages	Unassigned	Highest	To Do	Unresolved	Sep 21, 2026	Oct 25, 2026
MEM-69 Initial UI Integration and Review	Unassigned	Highest	To Do	Unresolved	Oct 18, 2026	Oct 25, 2026
MEM-68 Friend Search and Requests Page	Unassigned	Medium	To Do	Unresolved	Oct 12, 2026	Oct 21, 2026
MEM-67 Vignette Display and Edit Page	Unassigned	Highest	To Do	Unresolved	Oct 12, 2026	Oct 20, 2026
MEM-66 Vignette Candidate Page	Unassigned	High	To Do	Unresolved	Oct 9, 2026	Oct 18, 2026
MEM-34 Albums Collection and Individual Album Pages	Unassigned	Medium	To Do	Unresolved	Oct 15, 2026	Oct 23, 2026
MEM-33 Scrapbook Page	Unassigned	Highest	To Do	Unresolved	Oct 12, 2026	Oct 18, 2026
MEM-32 Make a Post Page	Unassigned	Highest	To Do	Unresolved	Oct 5, 2026	Oct 16, 2026
MEM-31 Friends Page	Unassigned	Medium	To Do	Unresolved	Oct 12, 2026	Oct 21, 2026
MEM-30 Collage Page	Unassigned	High	To Do	Unresolved	Oct 5, 2026	Oct 13, 2026
MEM-29 Settings Page	Unassigned	High	To Do	Unresolved	Sep 28, 2026	Oct 9, 2026
MEM-28 Profile Page	Unassigned	Medium	To Do	Unresolved	Sep 28, 2026	Oct 7, 2026
MEM-27 Onboarding/Registration Page	Unassigned	Highest	To Do	Unresolved	Sep 21, 2026	Oct 4, 2026

Work	Assignee	Priority	Status	Resolution	Start date	Due date
MEM-73 Back-End Development	Unassigned	Highest	To Do	Unresolved	Sep 21, 2026	Nov 16, 2026
MEM-84 Configure Secrets and Environment Variables	Unassigned	High	To Do	Unresolved	Sep 24, 2026	Oct 2, 2026
MEM-71 Initialize Backend Project	Unassigned	Highest	To Do	Unresolved	Sep 21, 2026	Sep 27, 2026
MEM-37 API development	Unassigned	High	To Do	Unresolved	Sep 28, 2026	Nov 8, 2026
MEM-93 Open-Meteo API Setup	Unassigned	Medium	To Do	Unresolved	Oct 1, 2026	Oct 7, 2026
MEM-70 SendGrid API Setup	Unassigned	High	To Do	Unresolved	Oct 1, 2026	Oct 7, 2026
MEM-65 Document and Test API in SwaggerHub	Unassigned	Medium	To Do	Unresolved	Oct 1, 2026	Nov 8, 2026
MEM-64 Implement Error Handling	Unassigned	High	To Do	Unresolved	Oct 1, 2026	Nov 1, 2026
MEM-63 Media Upload API	Unassigned	High	To Do	Unresolved	Oct 7, 2026	Oct 16, 2026
MEM-49 Settings API	Unassigned	Medium	To Do	Unresolved	Oct 5, 2026	Oct 9, 2026
MEM-48 Album API	Unassigned	Medium	To Do	Unresolved	Oct 16, 2026	Oct 30, 2026
MEM-47 Friends API	Unassigned	Medium	To Do	Unresolved	Oct 12, 2026	Oct 23, 2026
MEM-46 Collage API	Unassigned	High	To Do	Unresolved	Oct 12, 2026	Oct 20, 2026
MEM-45 Scrapbook API	Unassigned	Highest	To Do	Unresolved	Oct 5, 2026	Oct 16, 2026
MEM-44 Profile API	Unassigned	Medium	To Do	Unresolved	Oct 1, 2026	Oct 7, 2026
MEM-38 Authentication API	Unassigned	Highest	To Do	Unresolved	Sep 28, 2026	Oct 4, 2026

Work	Assignee	Priority	Status	Resolution	Start date	Due date
MEM-25 Database and Storage Setup	Unassigned	Highest	To Do	Unresolved	Sep 21, 2026	Sep 27, 2026
MEM-62 Input Test Data in Tables	Unassigned	Low	To Do	Unresolved	Sep 26, 2026	Sep 27, 2026
MEM-61 Create Photos Table	Unassigned	High	To Do	Unresolved	Sep 25, 2026	Sep 27, 2026
MEM-60 Create Album Table	Unassigned	Medium	To Do	Unresolved	Sep 25, 2026	Sep 27, 2026
MEM-59 Create Friends Table	Unassigned	Medium	To Do	Unresolved	Sep 24, 2026	Sep 26, 2026
MEM-58 Create Settings Table	Unassigned	Medium	To Do	Unresolved	Sep 24, 2026	Sep 26, 2026
MEM-57 Create Accounts and Authentication Tables	Unassigned	Highest	To Do	Unresolved	Sep 23, 2026	Sep 24, 2026
MEM-56 Configure On-device Local Storage	Unassigned	Highest	To Do	Unresolved	Sep 21, 2026	Sep 24, 2026
MEM-43 Create On-device Visited Areas Storage	Unassigned	High	To Do	Unresolved	Sep 24, 2026	Sep 27, 2026
MEM-42 Create Scrapbook Table	Unassigned	Highest	To Do	Unresolved	Sep 24, 2026	Sep 26, 2026
MEM-41 Create On-device Vignette Candidate Storage	Unassigned	Highest	To Do	Unresolved	Sep 23, 2026	Sep 26, 2026
MEM-40 Create Profiles Table	Unassigned	Medium	To Do	Unresolved	Sep 23, 2026	Sep 24, 2026
MEM-39 Create the Database	Unassigned	Highest	To Do	Unresolved	Sep 21, 2026	Sep 24, 2026
MEM-72 On-device Context and Moment Detection	Unassigned	Highest	To Do	Unresolved	Sep 28, 2026	Nov 15, 2026
MEM-75 Implement Permission State Management	Unassigned	Highest	To Do	Unresolved	Sep 28, 2026	Oct 9, 2026
MEM-76 Implement Routine Zone Storage and Matching	Unassigned	High	To Do	Unresolved	Oct 12, 2026	Oct 23, 2026
MEM-77 Implement Location Sampling	Unassigned	Highest	To Do	Unresolved	Oct 26, 2026	Nov 4, 2026
MEM-78 Implement Storing of Visited-area Data	Unassigned	High	To Do	Unresolved	Oct 29, 2026	Nov 8, 2026
MEM-79 Integrate Weather Context	Unassigned	Medium	To Do	Unresolved	Nov 2, 2026	Nov 8, 2026
MEM-80 Implement Activity and Motion Recognition	Unassigned	High	To Do	Unresolved	Nov 5, 2026	Nov 11, 2026
MEM-81 Implement Moment Scoring	Unassigned	Highest	To Do	Unresolved	Nov 8, 2026	Nov 13, 2026
MEM-82 Merge Related Candidates and Expire Old Data	Unassigned	High	To Do	Unresolved	Nov 10, 2026	Nov 15, 2026
MEM-74 Documentation	Unassigned	Medium	To Do	Unresolved	Sep 21, 2026	Nov 16, 2026